MuMMy: Multimodal Dataset supporting VLM-based Egyptology Research Assistant

Maksim Golyadkin
AIRI
Moscow, Russian Federation
mgolyadkin@hse.ru

Innokentiy Humonen
AIRI
Moscow, Russian Federation
humonen@airi.net

Valeria Rubanova
ITMO University
Saint Petersburg, Russian Federation
467271@niuitmo.ru

Danil Kalin
ITMO University
Saint Petersburg, Russian Federation
kalindan12@mail.ru

Ianis Plevokas
HSE University
Moscow, Russian Federation
yaaplevokas@edu.hse.ru

Dmitry Nikolotov
MIPT
Dolgoprudny, Russian Federation
nikolotov.dv@phystech.edu

Alexander Utkov
ITMO University
Saint Petersburg, Russian Federation
412515@niuitmo.ru

Nikita Sidelnikov
ITMO University
Saint Petersburg, Russian Federation
nsidelnikov@itmo.ru

Petr Ivanov
ITMO University
Saint Petersburg, Russian Federation
petr_ivanov@niuitmo.ru

Ekaterina Bureeva
HSE University
Moscow, Russian Federation
eibureeva@edu.hse.ru

Ekaterina Alexandrova
HSE University
Moscow, Russian Federation
ev.aleksandrova@hse.ru

Ilya Makarov
AIRI
Moscow, Russian Federation
ISP RAS
Moscow, Russian Federation
ITMO University
Saint Petersburg, Russian Federation
iamakarov@hse.ru

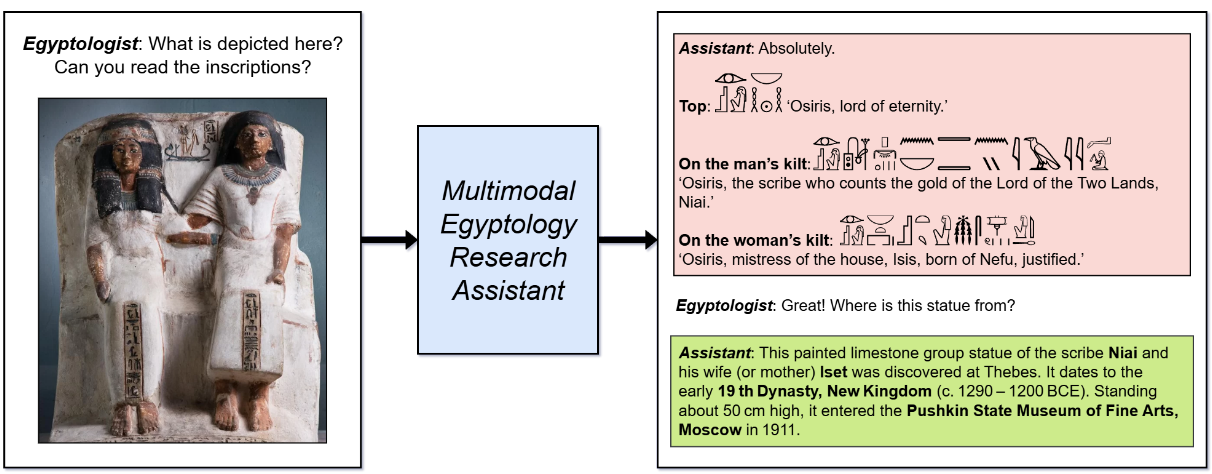


Figure 1: Example dialogue between an Egyptologist and an AI assistant. The green-highlighted response represents a task current systems can handle (tested on ChatGPT o3), while the red-highlighted response illustrates a task beyond current capabilities. The proposed dataset is aimed to bridge this gap.

MM '25, Dublin, Ireland

Abstract

We present the first multimodal dataset **MuMMy**, for developing research assistants that can interpret Egyptian hieroglyphic texts.

ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3758231

It pairs images with Gardiner codes, transliteration, and English translation at two levels of granularity. We also evaluate several deep learning pipelines across OCR, transliteration, and translation tasks, revealing the complexity of the domain and the challenges posed by error accumulation.

CCS Concepts

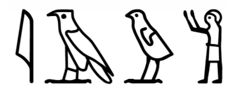
• **Computing methodologies** → **Computer vision**; **Machine translation**; • **Applied computing** → *Language translation.*

Keywords

multimodal dataset, ocr, machine translation, research assistant

ACM Reference Format:

Maksim Golyadkin, Innokentiy Humonen, Valeria Rubanova, Danil Kalin, Ianis Plevokas, Dmitry Nikolotov, Alexander Utkov, Nikita Sidelnikov, Petr Ivanov, Ekaterina Bureeva, Ekaterina Alexandrova, and Ilya Makarov. 2025. MuMMy: Multimodal Dataset supporting VLM-based Egyptology Research Assistant. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland*. ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3746027.3758231



	Code	English
ChatGPT 4o	A-G1-G43-A1	to see
ChatGPT o3	M17-G1-G43:A28	jubilation
Deepseek R1	-	to attack
Claude 3.7	T30-G5-G43-A28	praise
Ground Truth	M17-G1-G43-A30	praise

Table 1: Empirical demonstration of current LLMs' inability to accurately process Egyptian hieroglyphs. Models were asked to map hieroglyphs to Gardiner codes and translate them into English. Since recognition failed consistently, we supplied the correct codes to test translation. Even common word is difficult highlighting the need for fine-tuning.

1 Introduction

Studying Ancient Egyptian is both fascinating and important, offering insights into the legacy of one of the world's oldest civilizations. However, research in this area remains difficult: it requires deep interdisciplinary knowledge and access to vast amounts of data, much of which is still undigitized. While digitized texts can be processed in seconds using modern tools, Egyptologists often rely on physical books or scanned copies, working in a slow, manual fashion that has changed little in decades.

Recent advances in large language models, which are capable of reasoning [15, 23], enhanced with retrieval-augmented generation [12, 16] and multimodal inputs [17, 18], provide a solid foundation for creating AI assistants that could simplify Egyptology research (see Figure 1). Some LLMs exhibit a modest amount of knowledge about Ancient Egyptian from their broad training data. However, when rigorously tested, they often hallucinate even on simple tasks involving ancient texts, as illustrated in Table 1. Models frequently produce confident but completely wrong translations, misidentify signs, or invent hieroglyphs that don't exist, which highlights the need for fine-tuning to make them reliable tools for serious research.

A key barrier to fine-tuning is the lack of annotated data. Existing image-based datasets [8, 10, 11] primarily address hieroglyph recognition, with little emphasis on understanding the meaning of continuous texts. Datasets covering the textual modality, such as [1], are based on nearly all digitized material available today but remain small by modern deep learning standards. Finally, no multimodal datasets exist that link images with textual annotations, making it impossible to train end-to-end systems capable of processing Egyptian texts.

In this work, we introduce the first multimodal dataset **MuMMy** (**Mu**ltimodal **M**anuscript & **M**onument Repositor**y**), that bridges the gap between image and text modalities for Egyptian hieroglyphic texts. Each data point is a tuple consisting of an image of hieroglyphic writing, its corresponding Gardiner codes, transliteration, and English translation. The dataset is structured at two levels of granularity (clause-level and paragraph-level) resulting in 1090 annotated segments drawn from 87 source images. Textual annotations were sourced from the Thesaurus Linguae Aegyptiae, the most comprehensive collection of digitized Ancient Egyptian texts. For each selected text, we manually identified copyright-free image sources. While we do not introduce new texts, we are the first to align these existing resources across modalities. The dataset will be released as open-source under the permissive CC BY-SA 4.0 license and made publicly available through popular platforms such as GitHub and Hugging Face to facilitate future research.

To evaluate the complexity of our dataset and establish strong baselines for future work with LLMs, we tested several pipelines combining standard deep learning models. Each pipeline covered the full sequence of tasks: OCR, transliteration, and translation. Despite the limited dataset size, results show reasonable performance across all stages, with clear degradation due to error accumulation. These findings confirm the difficulty of the task and underscore the need for systems capable of generalization and step-by-step reasoning to arrive at reliable interpretations.

As a result, we make the following contributions:

- We present the **first multimodal dataset** for Egyptian hieroglyphic texts.
- We provide detailed hieroglyph-level **OCR annotations**.
- We **evaluate** multiple deep learning pipelines covering OCR, transliteration, and translation tasks.

2 Related Works

Previous work on Egyptian hieroglyphs treats vision and language tasks separately. In computer vision, research is limited to symbol recognition [3, 8–10]. In NLP, sequence-to-sequence models have been trained on Thesaurus Linguae Aegyptiae corpora for transliteration and translation only [4, 24], assuming clean text as input. Although one study combines OCR and translation [22], it does not evaluate the pipeline end-to-end. Recently, the MEH [11] dataset was introduced to map images of hieroglyphic sequences to their corresponding Gardiner codes, focusing solely on transcription of the script's visual form and omitting semantic annotations such as transliterations or translations.

Table 2: Comparison with existing hieroglyph datasets. MuMMy is the largest and the first to provide multimodal annotations.

Dataset	Images	Classification	OCR	Multimodal
Glyph [8]	6	✓	×	×
MMM [10]	19	✓	×	×
MEH [11]	40	✓	✓	×
MuMMy	87	✓	✓	✓

Our dataset brings these efforts together by unifying images with Gardiner codes, transliteration, and English translation, enabling true end-to-end evaluation of OCR-to-translation systems and supporting future multimodal research assistants. As shown in Table 2, MuMMy is the largest dataset of its kind and the first to unify images, Gardiner codes, transliterations, and translations, enabling complete multimodal processing of hieroglyphic texts from image to meaning.

3 Dataset

MuMMy is a multimodal dataset designed to bridge visual and linguistic research on Ancient Egyptian texts. It captures hieroglyphic inscriptions at multiple levels of granularity and connects them to expert-verified transliterations and translations. This section outlines the dataset's structure and licensing, describes the construction process and annotation pipeline, and presents key statistics that characterize its content.

3.1 Creation Process

Transliterating and translating hieroglyphic texts is a core part of Egyptological research and requires years of training. Creating such annotations from scratch would be prohibitively slow. Instead, we leveraged the largest available collection of expert-curated, digitized Egyptian texts: the Thesaurus Linguae Aegyptiae (TLA) [20], released under a permissive CC BY-SA 4.0 International license.

TLA texts are divided into clauses and include detailed annotations: Gardiner codes (a standardized symbolic encoding of hieroglyphs), transliteration, and translations (primarily in German, with a subset in English). Since English translations are essential for broader applicability, we restricted our dataset to texts with English translations only.

TLA provides textual data only, with no associated images. To create a multimodal resource, we first identified **all available** TLA texts with English translations and located corresponding images. These were collected from museum archives (which are copyright-free for research use) and from historical publications now in the public domain due to age. We collected 47 new images and complemented them with 40 images from MEH [11]. For the MEH subset, we reused the original OCR labels and enriched them with multimodal annotations.

Annotators manually (**no automated tools were used**) linked image regions to the corresponding TLA clauses (see Figure 2). For each clause, they drew a polygon around the relevant part of the image and assigned Gardiner codes, transliteration, and English translation from the digitized TLA text. Using the provided translation and the visual layout of the inscriptions, they grouped clauses into paragraphs. Paragraph regions were constructed by merging clause polygons into a unified mask with internal holes removed, and the text content was ordered using clause positions within each paragraph. In parallel, annotators created OCR-style annotations by segmenting individual hieroglyphs using bounding boxes and polygons, assigning each a Gardiner code. Rows and columns of text were also marked to support layout-aware recognition models.

To ensure label quality, annotators rendered the Gardiner codes using digital hieroglyph fonts and visually checked that the result matched the corresponding inscription in the image. Clauses with mismatches or uncertain readings were excluded. A PhD-level Egyptologist conducted a final pass over the dataset to validate the annotations.

TLA texts vary in editorial conventions across contributors. We applied preprocessing to standardize transliteration and translation style, removing uncertainty markers (e.g., (?), (!)) and editorial symbols that would not be expected in model outputs.

The annotation process was carried out by five Egyptology students over the course of a university practicum. The average time to annotate a single image was 78 minutes. Annotation was done without financial compensation.

3.2 Structure

MuMMy consists of paired visual and textual data at two levels of granularity: clauses and paragraphs. Each data point includes four elements: an image of the hieroglyphic text (cropped from a full source image via polygon annotation), its Gardiner code sequence (a symbolic encoding of individual hieroglyphs), a transliteration (a specialized Latin-based script used in Egyptology), and an English translation.

Since real-world inputs to a research assistant are more likely to be full paragraphs, this level serves as the most realistic setting for model evaluation. However, current models perform poorly at this scale. To enable tractable training and more targeted diagnostics, we also include clause-level samples: short, self-contained segments directly aligned with the structure of the TLA. This two-level design supports controlled benchmarking: clauses offer a manageable starting point for current models, while paragraphs challenge systems to handle longer texts.

To support OCR-specific workflows, we provide separate annotation files on full images. Each hieroglyph is labeled with its Gardiner code and spatial region (bounding box or polygon), along with row and column structure. This setup allows researchers to flexibly extract training samples suited to their own pipelines: be it for classification, detection, or full-layout OCR, without being tied to clause or paragraph definitions.

We do not prescribe a fixed data split. Users are encouraged to adapt the dataset to their own goals. However, for our baseline experiments, we include a predefined train/test split using a 1:2 ratio based on source images. All images from a given source (e.g., the same museum artifact or publication) are assigned entirely to either the training or test set. This design simulates out-of-domain generalization, a core challenge in Egyptology where every stele, statue, or inscription presents unique visual characteristics. Clauses and paragraphs inherit their split from the source image they were cropped from.

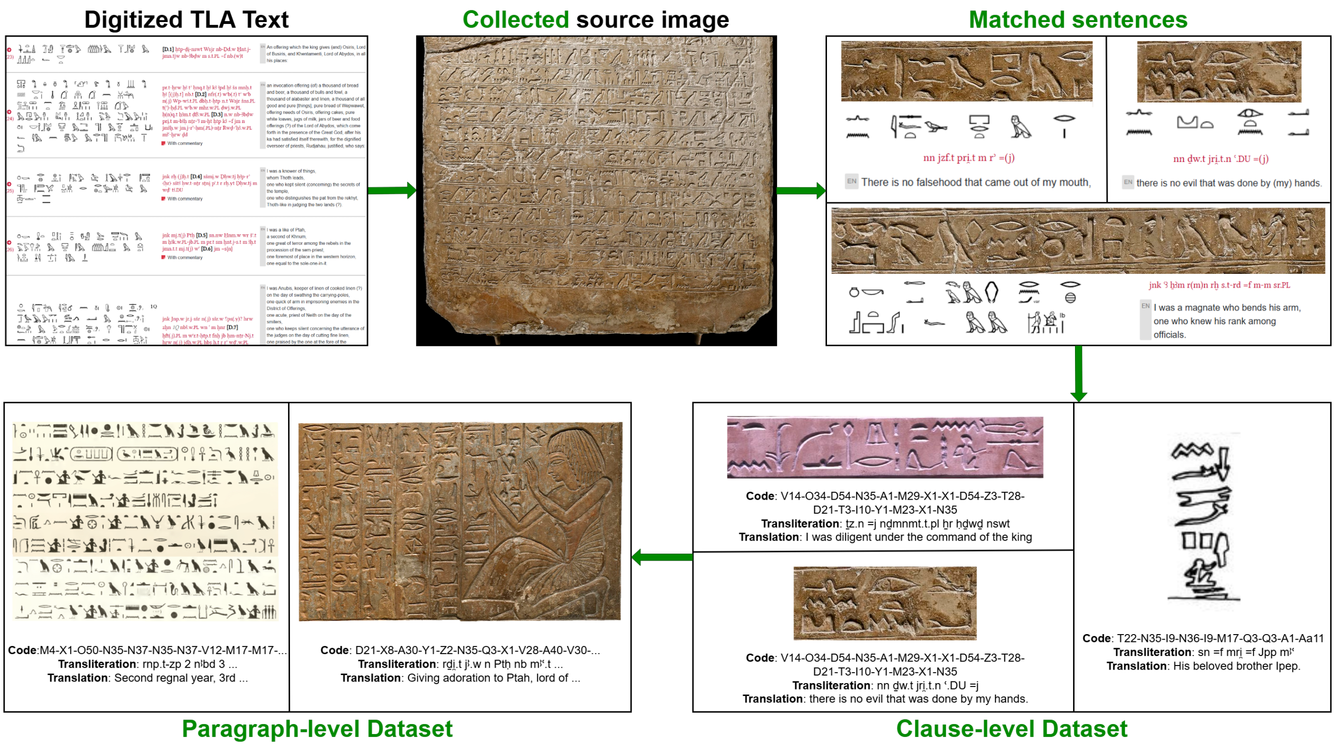


Figure 2: Dataset creation and structure. Textual data with English translations is sourced from the Thesaurus Linguae Aegyptiae (TLA) and matched with copyright-free images of original inscriptions. Annotators identify and align the corresponding regions in the images, creating clause- and paragraph-level samples. This process ensures high annotation quality, while the dual-level structure enables detailed analysis of model performance and failure cases. Green highlights our contributions.

Data is organized in a simple structure: images are stored as PNG files, and all annotations are stored in JSON format. The dataset is going to be distributed under the CC BY-SA 4.0 license, consistent with the licensing of the TLA texts. All accompanying images are either explicitly approved for academic reuse by museums or sourced from historical publications now in the public domain.

3.3 Statistics

The dataset contains a total of **87 source images**, from which we derive **984 clause-level samples** and **106 paragraph-level samples**. On average, clause-level samples contain **28 hieroglyphs**, **11 transliterated words**, and **18 English words** in translation. Paragraph-level samples are significantly longer, with **238 hieroglyphs**, **94 transliterated words**, and **156 English words** on average.

The OCR annotation layer spans the full source images and includes **26,486 individual hieroglyph instances**. These cover **704 distinct Gardiner codes**, approaching the complete Gardiner sign list (typically estimated at around 750 symbols, depending on variant inclusion). On average, each full image contains **304 hieroglyphs**.

In terms of source diversity, the dataset includes **32 unique physical artifacts** (e.g., stelae and inscribed stone blocks) and **55 images** sourced from historical publications. These publications represent **22 distinct sources**, meaning some books contribute multiple images. Every artifact is visually and stylistically distinct, emphasizing the dataset's heterogeneity and relevance for testing domain generalization.

4 Baseline

To demonstrate the applicability of the **MuMMy** dataset and establish initial performance benchmarks, we implement a full OCR-to-translation pipeline. We adopt the pipeline structure utilized in [9, 22], consisting of OCR, word segmentation, transliteration, and translation, and evaluate performance both for each component individually and in an end-to-end setting. In these experiments, our goal is not to achieve the quality expected of a practical research assistant, but to establish a baseline for future comparisons with vision-language models.

4.1 Component Datasets

To train and evaluate the individual components of our pipeline, we use several datasets, selected to maximize the amount of available training data. For OCR, we use the OCR annotation of **MuMMy**, described earlier. For textual components, we primarily rely on resources from the Thesaurus Linguae Aegyptiae (TLA).

Segmentation models are trained on the TLA 2018 dump [19], which we augmented with word boundary labels. For transliteration

Region Segmentation		Hieroglyph Detection		Hieroglyph Classification	
Model	**mAP**	**Model**	**mAP**	**Model**	**F1**
YOLO11s	0.64	YOLO11s	0.58	ResNet-50	0.65
YOLO11x	0.73	YOLO11x	0.71	Swin-B	0.84
DINO (ResNet-50)	0.56	DINO (ResNet-50)	0.43	Swin-L	0.89
DINO (Swin-L)	0.73	DINO (Swin-L)	0.69	**ConvNeXt-B**	**0.91**
CO-DETR (Swin-L)	**0.75**	**CO-DETR (Swin-L)**	**0.73**	ConvNeXt-L	0.85

Table 3: Performance comparison for OCR subtasks. Left: region segmentation models evaluated on row/column detection. Center: hieroglyph detection models evaluated on symbol localization. Detection scores are mAP@[0.5:0.95]. Right: classification models evaluated on cropped symbols using weighted F1. Results show that each subtask achieves high accuracy when evaluated independently, indicating that the performance bottlenecks arise only when the components are combined end-to-end.

model	precision	recall	F-score
MM	0.75	0.88	**0.81**
BERT	0.75	0.71	0.73
GU	0.60	0.89	0.72
all ones	0.32	1.00	0.49
all zeros	1.00	0.02	0.04

Table 4: The results of the word segmentation algorithms comparison. Names "all ones" and "all zeros" correspond to dummy segment-all and segment-nothing baselines. The non-ML MaxMatch algorithm outperforms the BERT-based model due to the limited amount of training data.

Input Type	BLEU ↑	ROUGE-L ↑
Hieroglyphs	29.97	0.5382
Transliteration	35.91	0.5996

Table 5: Results of automated translation evaluation. Transliteration-based inputs yield better scores due to the simpler nature of the task and greater availability of training data.

and translation, we constructed a unified dataset by combining and preprocessing two sources: (1) *TLA-Earlier* [1], which provides 12,800 clause-level triplets consisting of hieroglyphic text, standard transliteration, and German translation; and (2) *bbaw_egyptian* [24], which offers 35,500 sentence alignments with hieroglyphs, transcriptions, and translations in both German and English.

The split for OCR data follows the general **MuMMy** clause division used in end-to-end experiments. The textual datasets was randomly split into train/val/test sets using a 90-5-5 ratio.

4.2 OCR

Due to the limited availability of annotated data, we designed OCR system as a modular pipeline, where each model is responsible for a simple task. Under low-data conditions, this approach proved more effective than using fewer models to solve more complex problems.

The pipeline begins with region segmentation, where an object detection model identifies rows and columns of hieroglyphic text. For each detected region, a second hieroglyph detection model localizes individual symbols and distinguishes hieroglyphs from modern script. Detected hieroglyphs are then passed to an image classification model that predicts both the corresponding Gardiner code and the symbol's orientation. We use the predicted orientation of symbols within each region to inform a topological sorting algorithm, which reconstructs the correct reading order of hieroglyphs. The resulting sequence is used to generate the final recognition output.

For all subtasks, we trained and evaluated several pretrained architectures, selecting the best-performing models based on quality on test sets (see Table 3). The results show that models with stronger performance on their original pretraining tasks also perform better on our data, suggesting that images of Egyptian texts are not drastically out-of-domain and that fine-tuning provides substantial benefit.

4.3 Word Segmentation

The word segmentation problem involves dividing a continuous sequence of symbols into meaningful word units. It is an important task in NLP, particularly for languages lacking explicit word boundaries [21], OCR outputs, and ancient documents (e.g. *scriptio continua* [5]). In its turn, Egyptian presents distinct word segmentation challenges due to its logographic-phonographic writing system, lack of word spacing, and synthetic grammatical structure.

Following established practices [13], [2], we used neural network- and dictionary-based approaches, namely BERT [7] for sequence labeling, Max Match Word Segmenter (MM) [14] and its modification Greedy Unigram Word Segmenter (GU). MM greedily selects the longest dictionary-matched words sequentially, whereas GU picks the highest-frequency unigrams at each step; both relying on preloaded lexical data (a dictionary or unigram frequencies, respectively). Word frequencies for GU were computed on TLA corpus ([20]), and two dictionaries was tested as MM backbones, one also obtained from TLA corpus and another gained from the [1]. Comparison of segmentation algorithms presented in Table 4. Due to lower scores, GU was excluded from the end-to-end pipeline evaluation.

4.4 Transliteration

For transliteration, we use a simple yet effective baseline: a random forest classifier trained on a bag-of-hieroglyphs representation at the word level. This model effectively maps each hieroglyphic word

Pipeline Config			OCR	Transliteration				Translation (EN)			
Seg	T-lit	T-late	CER ↓	BLEU ↑	ROUGE-L ↑	METEOR ↑	ChrF ↑	BLEU ↑	ROUGE-L ↑	METEOR ↑	ChrF ↑
MM	RF	H	0.54/1.02	7.95/**14.3**	0.38/**0.42**	**0.20**/**0.22**	**27.0**/**33.5**	4.43/4.74	0.22/0.20	**0.25**/**0.20**	28.0/23.4
MM	RF	T	0.54/1.02	7.95/**14.3**	0.38/**0.42**	**0.20**/**0.22**	**27.0**/**33.5**	2.29/3.45	0.15/0.15	0.22/0.19	25.7/**26.5**
MM	TR	H	0.54/1.02	6.66/3.30	0.34/0.29	0.11/0.08	16.1/11.1	4.43/4.74	0.22/0.20	**0.25**/**0.20**	28.0/23.4
MM	TR	T	0.54/1.02	6.66/3.30	0.34/0.29	0.11/0.08	16.1/11.1	2.20/2.75	0.15/0.15	0.22/0.18	24.7/24.0
BERT	RF	H	0.54/1.02	**11.3**/6.52	0.39/0.37	0.18/0.15	24.8/21.3	5.38/**4.83**	0.23/**0.22**	**0.25**/**0.20**	**28.4**/24.9
BERT	RF	T	0.54/1.02	**11.3**/6.52	0.39/0.37	0.18/0.15	24.8/21.3	**11.3**/2.86	**0.39**/0.16	0.18/0.17	24.8/24.1
BERT	TR	H	0.54/1.02	7.65/3.69	0.35/0.30	0.11/0.09	15.6/11.6	5.38/**4.83**	0.23/**0.22**	**0.25**/**0.20**	**28.4**/24.9
BERT	TR	T	0.54/1.02	7.65/3.69	0.35/0.30	0.11/0.09	15.6/11.6	3.09/3.48	0.17/0.17	0.22/0.19	26.2/25.3
None	TR*	H*	0.54/1.02	7.68/4.63	**0.42**/0.35	0.15/0.11	19.6/13.6	3.68/2.55	0.17/0.18	0.17/0.16	23.8/19.8
None	TR*	T*	0.54/1.02	7.68/4.63	**0.42**/0.35	0.15/0.11	19.6/13.6	2.23/1.90	0.15/0.13	0.20/0.16	24.2/23.1

Table 6: End-to-end pipeline evaluation (clause / paragraph). Seg: segmentation method (MM = MaxMatch, BERT = BERT-based, None = no segmentation). T-lit: transliteration model (RF = Random Forest, TR = Transformer). T-late: representation fed to the translator (H = hieroglyphs, T = transliteration). An asterisk (*) indicates input without spaces. Results show limited overall performance and discrepancies across metrics, underscoring the difficulty of the task and the need for more robust models.

to its most frequently observed transliteration, which is often sufficient, as frequent words tend to have a dominant transliteration. Additionally, we fine-tune the NLLB model [6] for transliteration in a standard machine translation setup.

4.5 Translation

To evaluate the translation component of our pipeline, we fine-tune the *facebook/nllb-200-distilled-600M* model [6], a state-of-the-art multilingual NMT system covering over 200 languages. The Egyptian language is low-resource by modern standards, motivating the use of a pretrained multilingual model. Training followed standard NMT practices, including a linear learning rate scheduler and gradient accumulation.

We assess translation quality using two common automated metrics: BLEU and ROUGE-L, in line with prior work [4]. Table 5 presents the results for models translating from either hieroglyphs or transliteration.

4.6 End-to-end Pipeline Evaluation

To assess the real-world applicability of the proposed pipeline, we evaluate the whole pipeline in a end-to-end setting where each module receives the output of the previous stage, and errors naturally accumulate throughout the process. We choose ten pipeline configurations to analyze how different component choices affect overall performance. These variants differ in segmentation strategy (MaxMatch, BERT-based, or none), transliteration method (random forest or transformer model), and the representation used as input to the translation module (hieroglyphs or transliteration).

Evaluation is performed at three output levels: OCR is scored using Character Error Rate (CER) on predicted Gardiner codes, while transliteration and translation are evaluated with BLEU, ROUGE-L, METEOR, and ChrF.

Results, presented in Table 6, show that all components suffer from limited training data. Best performance is achieved with either MaxMatch or BERT-based segmentation, both of which consistently outperform the no-segmentation setting. RF-based transliteration proves superior to the transformer. Translating directly from hieroglyphs yields better results by avoiding error accumulation from the transliteration model.

While OCR subtasks individually achieve reasonable performance, end-to-end output remains weak due to error accumulation at every stage. In particular, early-stage region segmentation errors often lead to cascading failures. OCR remains the most data-constrained module, and its weaknesses limit the entire pipeline. These findings highlight the urgent need for expanding annotated data on the image modality.

5 Limitations and Future Work

The primary limitation of our work lies in the scarcity of annotated training data, particularly for long-tail hieroglyphs. This severely constrains model generalization and affects overall pipeline robustness. Additionally, English translations in our dataset are often short or derived from German, limiting their linguistic richness.

Future work will focus on expanding the dataset through both real and synthetic data sources. Our current pipeline already enables semi-automated annotation, laying the groundwork for human-in-the-loop workflows to scale data creation efficiently. We also plan to extend **MuMMy** to support additional target languages. Finally, evaluating and fine-tuning unified multimodal models (e.g., LLaVA) on this dataset is a promising next step toward building practical research assistants for Egyptology.

We invite Egyptologists and researchers working on other ancient languages to collaborate by using our tools, sharing data and expertise, and collectively advancing the interpretation of hieroglyphic texts and artifacts, along with related research.

6 Conclusion

We introduced **MuMMy**, the first multimodal dataset aligning images of Egyptian hieroglyphic texts with their transliterations and translations. Our benchmarks demonstrate the challenges of OCR-to-translation pipelines under limited data and highlight key bottlenecks across components. **MuMMy** lays the foundation for future research on multimodal Egyptology tools and vision-language modeling in low-resource historical domains.

Acknowledgements

We thank the AIRI development team led by Mamontov S. and the annotators from HSE: Viktoriya Fedorova, Elizaveta Frolova, and Albina Tsipileva. This work was supported by a grant, provided by the Ministry of Economic Development of the RF in accordance with the subsidy agreement (agreement identifier 000000C313925P4G0002) and the agreement with the Ivannikov Institute for System Programming of the Russian Academy of Sciences No. 139-15-2025-011.

References

[1] 2024. Thesaurus Linguae Aegyptiae, Original Earlier Egyptian sentences, corpus v18, premium. https://huggingface.co/datasets/thesaurus-linguae-aegyptiae/tla-Earlier_Egyptian_original-v18-premium

[2] Andrea Barucci, Chiara Canfailla, Costanza Cucci, Matteo Forasassi, Massimiliano Franci, Guido Guarducci, Tommaso Guidi, Marco Loschiavo, Marcello Picollo, Roberto Pini, et al. 2022. Ancient Egyptian hieroglyphs segmentation and classification with convolutional neural networks. In *International Conference Florence Heri-Tech: the Future of Heritage Science and Technologies*. Springer, 126–139.

[3] Andrea Barucci, Costanza Cucci, Massimiliano Franci, Marco Loschiavo, and Fabrizio Argenti. 2021. A Deep Learning Approach to Ancient Egyptian Hieroglyphs Classification. *IEEE Access* 9 (2021), 123438–123447.

[4] Mattia Cao, Nicola De Cao, Angelo Colonna, and Alessandro Lenci. 2024. Deep Learning Meets Egyptology: a Hieroglyphic Transformer for Translating Ancient Egyptian. In *Proceedings of the 1st Workshop on Machine Learning for Ancient Languages (ML4AL 2024)*. 71–86.

[5] Thibault Clérice. 2020. Evaluating deep learning methods for word segmentation of Scripta Continua texts in old French and Latin. *Journal of Data Mining and Digital Humanities* 2020 (2020).

[6] Marta R Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, et al. 2022. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672* (2022).

[7] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.

[8] Morris Franken and Jan C. van Gemert. 2013. Automatic Egyptian hieroglyph recognition by retrieving images as texts. In *Proceedings of the 21st ACM International Conference on Multimedia* (Barcelona, Spain) *(MM '13)*. Association for Computing Machinery, New York, NY, USA, 765–768. doi:10.1145/2502081.2502199

[9] Maksim Golyadkin, Innokentiy Humonen, Yanis Plevokas, Ekaterina Bureeva, Ekaterina Alexandrova, and Ilya Makarov. 2025. Automatic Interpretation of Ancient Egyptian Texts for Education and Research *(SIGGRAPH Posters '25)*. Association for Computing Machinery, New York, NY, USA, Article 39, 3 pages. doi:10.1145/3721250.3743025

[10] Maksim Golyadkin, Valeria Rubanova, Aleksandr Utkov, Dmitry Nikolotov, and Ilya Makarov. 2025. Evaluation of Egyptian Hieroglyph Classification Across Diverse Writing Styles. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 1–8.

[11] Maksim Golyadkin, Valeria Rubanova, Aleksandr Utkov, Dmitry Nikolotov, and Ilya Makarov. 2025. MEH: A Multi-Style Dataset and Toolkit for Advancing Egyptian Hieroglyph Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 1–8.

[12] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. 2020. Retrieval Augmented Language Model Pre-Training. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 119)*, Hal Daumé III and Aarti Singh (Eds.). PMLR, 3929–3938. https://proceedings.mlr.press/v119/guu20a.html

[13] Heidi Jauhiainen and Tommi Jauhiainen. 2023. Automatic Word Segmentation for Egyptian Hieroglyphic Texts.. In *DH*.

[14] Daniel Jurafsky and James H. Martin. 2009. *Speech and Language Processing (2Nd Edition)*. Prentice-Hall, Inc., Upper Saddle River, NJ, USA.

[15] Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large Language Models are Zero-Shot Reasoners. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 22199–22213. https://proceedings.neurips.cc/paper_files/paper/2022/file/8bb0d291acd4acf06ef112099c16f326-Paper-Conference.pdf

[16] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (Eds.), Vol. 33. Curran Associates, Inc., 9459–9474. https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf

[17] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual Instruction Tuning. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 34892–34916. https://proceedings.neurips.cc/paper_files/paper/2023/file/6dcf277ea32ce3288914faf369fe6de0-Paper-Conference.pdf

[18] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*, Marina Meila and Tong Zhang (Eds.). PMLR, 8748–8763. https://proceedings.mlr.press/v139/radford21a.html

[19] Tonio Sebastian Richter, Ingelore Hafemann, Hans-Werner Fischer-Elfert, and Peter Dils. 2018. *Teilauszug der Datenbank des Vorhabens" Strukturen und Transformationen des Wortschatzes der ägyptischen Sprache" vom Januar 2018*. Berlin-Brandenburgische Akademie der Wissenschaften.

[20] Tonio Sebastian Richter, Daniel A. Werning, Hans-Werner Fischer-Elfert, and Peter Dils. 2025. Thesaurus Linguae Aegyptiae. Online Resource. https://thesaurus-linguae-aegyptiae.de Corpus issue 19, Web app version 2.2.0, 11/5/2024, ed. by Tonio Sebastian Richter & Daniel A. Werning on behalf of the Berlin-Brandenburgische Akademie der Wissenschaften and Hans-Werner Fischer-Elfert & Peter Dils on behalf of the Sächsische Akademie der Wissenschaften zu Leipzig (accessed: 11.2.2025).

[21] Xinxin Shu, Junhui Wang, Xiaotong Shen, and Annie Qu. 2017. Word segmentation in Chinese language processing. *Statistics and its Interface* 10, 2 (2017), 165–173.

[22] Asmaa Sobhy, Mahmoud Helmy, Michael Khalil, Sarah Elmasry, Youtham Boules, and Nermin Negied. 2023. An AI Based Automatic Translator for Ancient Hieroglyphic Language—From Scanned Images to English Text. *IEEE Access* 11 (2023), 38796–38804. doi:10.1109/ACCESS.2023.3267981

[23] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 24824–24837. https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf

[24] Philipp Wiesenbach and Stefan Riezler. 2019. Multi-Task Modeling of Phonographic Languages: Translating Middle Egyptian Hieroglyphs. *Proceedings of the International Workshop on Spoken Language Translation* (2019). https://www.cl.uni-heidelberg.de/statnlpgroup/publications/IWSLT2019_v2.pdf